Homework 1

Name: Number: 2026.3.12

1. [50%] A manufacturer determines that the profit P (in dollars) obtained by producing and selling x units of Product 1 and y units of Product 2 is approximated by the model

$$P(x, y) = 8x + 10y - (0.001)(x^2 + xy + y^2) - 10000.$$

Find the production level that produces a maximum profit. What is the maximum profit?

2. [50%] Determine whether the set of points satisfying a quadratic inequality:

$$C = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_2 \geq x_1^2\}$$

is convex. Provide a formal proof or a counterexample.